



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

IS3233-1 – FUNDAMENTALS OF IMAGE PROCESSING

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **The process of converting an analog image into a digital image is called:**
 - A) Image sampling
 - B) Image quantization
 - C) Image acquisition
 - D) Image digitization
2. **What is the purpose of image quantization?**
 - A) To smooth an image
 - B) To convert an image to binary
 - C) To reduce the number of colors in an image
 - D) To enhance contrast
3. **Which gray level transformation is used to compress the dynamic range of an image?**
 - A) Negative transformation
 - B) Logarithmic transformation
 - C) Power-law transformation
 - D) Thresholding
4. **The purpose of a sharpening filter in spatial domain image processing is to:**
 - A) Remove artifacts
 - B) Reduce image noise
 - C) Smooth edges
 - D) Enhance fine details
5. **Which of the following is a disadvantage of using high-pass filters in image processing?**
 - A) They blur the image
 - B) They reduce image contrast
 - C) They emphasize noise
 - D) They compress the image
6. **What is the main advantage of using a median filter for noise reduction?**
 - A) It preserves edges while removing noise
 - B) It reduces the image resolution
 - C) It compresses the image data
 - D) It sharpens the image
7. **Which of the following is a type of image enhancement in the spatial domain?**
 - A) Fourier filtering
 - B) Histogram equalization
 - C) JPEG compression
 - D) DCT transformation
8. **What is the primary function of image enhancement techniques in the spatial domain?**
 - A) To reduce the size of an image
 - B) To improve the visual appearance of an image
 - C) To compress an image for storage
 - D) To convert an image to grayscale
9. **What is the primary purpose of homomorphic filtering in image processing?**
 - A) To reduce image noise
 - B) To enhance the contrast of an image
 - C) To combine illumination and reflectance components
 - D) To perform image segmentation

10. In the context of segmentation, what is thresholding?
- A) A technique to smooth an image B) A method for enhancing image contrast
C) A process of separating pixels based on intensity D) A way to compress an image
11. Which of the following techniques is used to detect discontinuities in an image?
- A) Edge detection B) Smoothing
C) Fourier Transform D) Histogram equalization
12. Which image segmentation technique uses the motion of objects as a key factor?
- A) Thresholding B) Edge linking
C) Motion segmentation D) Region-based segmentation
13. In image compression, what is meant by "compression ratio"?
- A) The ratio of the original image size to the compressed image size B) The number of bits per pixel in the image
C) The amount of noise reduced during compression D) The quality of the compressed image
14. Which of the following is not a method for error-free compression?
- A) Run-length encoding B) Huffman coding
C) Arithmetic coding D) JPEG compression
15. What is the key benefit of using Huffman coding in image compression?
- A) It improves image resolution B) It reduces image size by eliminating redundancy
C) It enhances image contrast D) It increases the color depth of an image
16. Which technique is most effective for segmenting images with overlapping objects?
- A) Thresholding B) Edge detection
C) Watershed segmentation D) Region growing
17. What is the main advantage of using lossy compression over lossless compression?
- A) Higher compression ratios B) Better image quality
C) Faster compression times D) No loss of data
18. Which compression technique is typically used for reducing the size of scanned documents and faxes?
- A) JPEG B) TIFF
C) CCITT D) GIF
19. Which compression method is typically used for images with large areas of uniform color?
- A) JPEG B) PNG
C) GIF D) BMP
20. Which of the following operations is used to remove small objects from an image while preserving the shape of larger objects?
- A) Dilation B) Erosion
C) Opening D) Closing

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the components of an image processing system with a suitable block diagram.	10	L*2	1	1
b)	Illustrate histogram matching with an example	6	L2	2	1

2. a) Consider an image segment
 3 4 1 2 0
 0 1 0 4 2(Q)
 2 2 3 1 4
 (P)3 0 4 2 1
 Let $V=\{0,1,2\}$.Evaluate the length for shortest 4,8 and m path between P and Q . Repeat for $V=\{2,3,4\}$ 8 L3 2 2
 b) Explain image acquisition process using single sensor with a neat diagram. 8 L2 1 1
3. a) Explain image smoothing filters in spatial domain with a relevant equation. 9 L2 3 1
 b) Illustrate 1D Fourier Transform of one variable. 7 L2 3 1

Unit – II

4. a) What is thresholding in image processing? Describe the steps involved in determining a global threshold value (T). 8 L2 3 1
 b) Explain image smoothing filters in frequency domain with a relevant equation. 8 L2 4 1
5. a) Explain homomorphic filtering in frequency domain. 10 L2 3 1
 b) What is the Canny edge detection technique? Describe the main steps involved in this method. 6 L2 4 1
6. a) What is Image Compression? Describe the general process of an image compression system and include a clear diagram to illustrate the steps. 10 L2 4 1
 b) Explain LZW coding technique in detail. 6 L2 4 1

Unit – III

7. a) What are the characteristics used to differentiate between colors? Explain each characteristic. 10 L2 5 1
 b) Outline hit or mass transformation used for shape detection. 6 L2 5 1
8. a) Summarize CMY and CMK color models. 6 L2 5 1
 b) Illustrate erosion and dilation operation in morphological image processing. 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
